Radical equations

1 Solve the following equations:

a $\sqrt{x} = \sqrt{13}$ **b** $\sqrt{h} = 5$ **c** $\sqrt{x} = \frac{3}{8}$ **d** $-3\sqrt{k} = -12$

2 Define a radical equation.

3 Is 4 a solution of the equation $\sqrt{x} = -2$?

4 Is 3 a solution of the equation $\sqrt{x-2} - 4 = \sqrt{x+6}$?

5 Is -2 a solution of the equation $\sqrt{x+18} = x + 6$?

6 Solve the following equations:

a $\sqrt{y} - 6 = 1$

b $5\sqrt{x} + 3 = 28$

c $\sqrt{p} + 4 = 7$

d $\sqrt{y} + \frac{1}{5} = \frac{4}{5}$

e $\sqrt{x} - \frac{1}{4} = \frac{1}{3}$

f $\frac{\sqrt{x}}{-2} = -4$

g $5\sqrt{y} + 4 = 14$

h $-5\sqrt{c} - 2 = -17$

i $2\left(\sqrt{b} - 3\right) = 4$

j $\frac{\sqrt{a} - 1}{2} = 1$

k $\frac{\sqrt{b} + 1}{2} = \frac{3}{5}$

l $5\left(3\sqrt{a} + 8\right) = 70$

7 Solve the following equations:

a $\sqrt{x-5} = \sqrt{25}$

b $\sqrt{2x-6} = 2$

c $\sqrt{x-5} = 7$

d $\sqrt{x^2 - 8x + 16} = 7$

e $\sqrt{9x^2 - 48x + 64} = x$

f $\frac{12}{\sqrt{x-2}} = 3$

g $8\sqrt{x} - 4\sqrt{x} = 16$

h $\sqrt{3x-2} = 8$

i $\frac{\sqrt{x+9}}{2} - 8 = -2$

j $\frac{7\sqrt{x}}{11} + \frac{2\sqrt{x}}{11} = 9$

k $\sqrt{x^2 + 20} = 6$

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8 Ray is given the equation $\sqrt{x+3}+7=4$ to solve. His work is shown:

1. $\sqrt{x+3}+7=4$
2. $\sqrt{x+3}=-3$
3. $x+3=9$
4. $x=6$

a Does $x=6$ satisfy the equation $\sqrt{x+3}+7=4$?

b Describe the error Ray made in his work.

9 Tom and Katrina are both given the equation $2\sqrt{x+71}-10=8$ to solve. Both have made errors in their work below:

Tom's work

1. $2\sqrt{x+71}-10=8$
2. $\sqrt{x+71}-10=4$
3. $\sqrt{x+71}=14$
4. $x+71=196$
5. $x=125$

Katrina's work

1. $2\sqrt{x+71}-10=8$
2. $2\sqrt{x+71}=18$
3. $\sqrt{x+71}=9$
4. $x+71=3$
5. $x=-68$

a Describe Tom's error.

b Describe Katrina's error.

c Solve the equation $2\sqrt{x+71}-10=8$.

10 Consider the equation $\sqrt{x+3}+10=0$. Without solving, explain how we can see that the equation has no real solutions.

11 Consider the equation $10x=1+3\sqrt{x}$.

a Solve the equation for x.

b Which value of x is an extraneous solution?

12 Consider the equation $\sqrt{x-5}=\sqrt{x}-1$.

a Solve the equation for x.

b Is this value of x an extraneous solution?

13 Describe a correct first step to solve the equation $(x+7)^{\frac{3}{5}}=3$

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14 Solve the following equations:

a $\sqrt[3]{x+7}=4$ **b** $\sqrt[3]{3x-4}=2$ **c** $\sqrt[4]{x+2}+4=1$

15 Solve the following equations:

a $x^{\frac{3}{2}}=27$ **b** $x^{\frac{2}{7}}=4$ **c** $4x^{\frac{1}{4}}-3=9$ **d** $4x^{\frac{2}{7}}-1=15$

Applications

16 A square pyramid with a height of 12 units and a volume of V cubic units has a base with a side length of $b=\sqrt{\frac{V}{4}}$ units.

a Find the side length of the base of a square pyramid for each of the volumes in the table. Round your answers to the nearest tenth of a unit.

V	40	400	$4\,000$
b			

b Each volume in the table increased by a factor of 10. Did the corresponding side length of the base increase by a factor of 10?

17 An electron is constantly moving, and its position with respect to the fixed point ($x=0$) can be modeled by the equation $x=-(t-4)^2+4$, where x is its position after t seconds. After 4.5 seconds, it is at $x=3.75$.

a Solve the equation for t.

b Determine the time t at which the electron is at the fixed position.

c Determine how long it takes the electron to get from $x=-5.1$ to $x=-6.3$. Round your answer to the nearest tenth of a second.

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